

Working on Repository

-> We can perform below operations on Repository

- 1) Create Operation
- 2) Delete Operation
- 3) Clone Operation
- 4) Pull Operation
- 5) Push Operation
- 6) Rename Operation
- 7) Create branches on repository

-> Now, we will discuss each operation one by one.

-> We have two types of repositories

- 1) **Local Repository:** Present on your Local System.
- 2) **Central/Remote Repository:** Present on Server(github.com)

-> 1. Create Operation

- 1) If you want to create a new repository on local system then use "git init" command.

Once the command is executed then it will create a new empty repository.

At the time of repository creation one initial branch gets created and that branch name is either **main** or **master**.

After repository creation if we want to change initial branch name from master to main or main to master then use below command.

"git branch -M main" -> Changing master to main

"git branch -M master" -> Changing main to master

- 2) If we want to create remote repository (on github.com) then we follow below steps

Click on Profile Picture -> Your Organizations -> Repository Tab -> New Repository
or

Click on '+' Symbol -> New Repository

-> 2. Delete operation

1) If we want to delete a local repository then we use linux "rm" command

Syntax: "rm -r repository_name"

or right click on repository_name and delete

or

select repository_name and shift+delete

2) If we want to delete remote repository then follow the below steps

Select Desired Repository -> Settings -> Danger Zone -> Delete Repository

Note: But Real time we never delete Remote Repository ones it is created.

-> 3. Clone operation:

-> Cloning means creating duplicate copy.

-> Clone operation is used to create clone of central repository.

-> We create clone by using "git clone" command.

Syntax: git clone repository_url

-> When we clone a repository then by default main or master branch get cloned on local system.

Q. When we clone repository?

-> If remote repository copy is not available on your local system.

-> 4. pull operation

-> Pull operation is used to pull(download) the changes from remote repository to local repository.

-> To pull the changes follow the below steps

Open GitBash in local repository and simply type "git pull" and press enter.

-> Example: DAILY-NOTES repository

-> 5. push operation

-> push operation is used for two purpose

1) push newly created local repository on github.com

2) push local repository modification/changes to central repository.

1) push newly created local repository on github.com

-> If you want to push your newly created local repository on github then follow below steps.

- 1) Create new Local Repository.(say 'HRM')
- 2) Create new Repository on github with name same as local repository(HRM)
- 3) Local repository root branch name must be same as remote repository root branch name. If not same then rename the local repository root branch name by using below command.

`git branch -M main/master`

- 4) create link between local repository and remote repository by using command. "`git remote add origin REMOTE_URL`"
Above command add remote origin on local repository

- 5) execute `git push -u origin main/master` command

This is one time activity,next time onward if we want to push changes then simply use "`git push`" and press enter.

or If you want to pull changes then simply type "`git pull`" command

2) push local repository modification/changes to central repository.

-> If we want to push local repository changes on remote repository then open git bash and type "`git push`" and press enter.

-> Example: TASKS Repository.